Evaluation of the performance of grouted couplers when used for connections of reinforcing bars produced in Chile

- TEST REPORT -

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Tokyo Institute of Technology
Professor Junichiro Niwa

Professor Junichiro Niwa's profile

Personal Data:	Date of Birth : January 11, 1956 Place of Birth : Kanazawa, Ishikawa
Education:	1974－1978 : Department of Civil Engineering, The University of Tokyo, B. E. 1978－1980 : Master's Course, Graduate School of Civil Engineering, The University of Tokyo, M.E. 1980－1983 : Doctor's Course, Graduate School of Civil Engineering, The University of Tokyo, Dr. E.
Work Experience:	1983－1986 : Assistant Professor, The University of Tokyo 1986－1989 : Associate Professor, Yamanashi University 1989－1998 : Associate Professor, Nagoya University 1998 to date : Professor, Tokyo Institute of Technology
International Activity:	1990－1991 : Visiting Scholar at the University of Kansas, U.S.A. 1995－1997 : Associate Professor, Asian Institute of Technology, Bangkok (seconded by JICA) 2002.7－2002.8 : Japanese Language Teacher, Preparatory School in Northeast Normal, University, Changchun, China
Affiliations:	Japan Society of Civil Engineers (JSCE) Japan Concrete Institute (JCI) Japan Prestressed Concrete Institute (JPCI) International Association for Bridge and Structural Engineering (IABSE) International Federation for Structural Concrete (fib) American Concrete Institute (ACI)
Research Topics:	(1) Application of fracture mechanics of concrete to evaluate the size effect of structural concrete. (2) Localization phenomenon of compression failure of concrete. (3) Lattice model analysis for structural concrete subjected to static and/or dynamic loading. (4) Ultimate shear capacity of prestressed concrete beams. (5) Application of new materials, and/or new types of structures, such as prestressed concrete beams using pretensioned precast web, reinforced concrete members with steel fiber, super high strength cementitious composites with steel fiber, etc. (6) Repairing and strengthening method for structural concrete using sprayed fiber reinforced polymer. (7) Mechanical behavior of concrete using Pinatubo aggregate reinforced with palm fiber.
Selected Journal Papers: (2006 to date)	1. Bui, K.D., Niwa, J., "Ultimate Tendon Stress in Beams Prestressed with External Tendons", Journal of Materials, Concrete Structures and Pavements of JSCE (Doboku Gakkai Ronbunshuu E), Vol.62, No.1, 2006.3, pp.211－225. 2. Bui, K.D., Sivaleepunth, C., Niwa, J., "Slip Phenomenon at Deviators in Beams Prestressed with External Tendons", Journal of Materials, Concrete Structures and Pavements of JSCE (Doboku Gakkai Ronbunshuu E), Vol.62, No.1, 2006.3, pp.227－239. 3. Sivaleepunth, C., Niwa, J., Diep, B.K., Tamura, S., Hamada, Y., "Prediction of Tendon Stress and Flexural Strength of Externally Prestressed Concrete Beams", Journal of Materials, Concrete Structures and Pavements of JSCE (Doboku Gakkai Ronbunshuu E), Vol.62, No.1, 2006.3, pp.260－273. 4. Bui, K.D., Niwa,J., "Prediction of Loading-Induced Stress in Unbonded Tendons at Ultimate", Journal of Materials, Concrete Structures and Pavements of JSCE (Doboku Gakkai Ronbunshuu E), Vol. 62, No.2, 2006.6, pp.428－443.

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8. Sivaleepunth, C., Niwa, J., Nguyen, D. H., Hasegawa, T., Hamada, Y., “Shear Carrying Capacity of Segmental Prestressed Concrete Beams”, Journal of Materials, Concrete Structures and Pavements of JSCE (Doboku Gakkai Ronbunshuu E), Vol. 65, No. 1, 2009.1, pp.63－75.

9. Nguyen, D.H., Watanabe, K., Niwa, J., Hasegawa, T., "Modified Model for Shear Carrying Capacity of Segmental Concrete Beams with External Tendons", Journal of Materials, Concrete Structures and Pavements of JSCE (Doboku Gakkai Ronbunshuu E), Vol. 66, No. 1, 2010. 1, pp.53－67.

10. Watanabe, K., Kimura, T. and Niwa, J., "Synergetic effect of steel fibers and shear reinforcing bars on the shear-resistance mechanisms of RC linear members", Journal of Construction and Building Materials, Elsevier, Vol.24, No.12, 2010.12, pp.2369－2375.

11. Jongvivatsakul, P., Watanabe, K., Matsumoto,K., Niwa, J., “Evaluation of Shear Carried by Steel Fibers of Reinforced Concrete Beams Using Tension Softening Curves”, Journal of Materials, Concrete Structures and Pavements of JSCE (Doboku Gakkai Ronbunshuu E), Vol. 67, No. 4, 2011.10, pp.493－507.

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16. Wirojjanapirom, P., Matsumoto, K., Kono, K., Niwa, J., “Experimental Study on Shear Behavior of RC Beams Using U-Shaped UFC Permanent Formwork with Shear Keys and Bolts”, Journal of Materials, Concrete Structures and Pavements of JSCE (E2), Vol. 69, No. 1, 2013, pp.67－81.

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1. Objective of the tests

NMB Splice Sleeve model UX has been tested and evaluated in Japan and US by approved institutions and received from them Quality Certifications or Evaluation Reports. These documents are necessary for Splice Sleeve users in these countries to submit to local authorities and get approval for its use. By the other hand, there are countries where in most cases it is required to conduct performance tests using local reinforcing bars.

This is the report of a series of tests that were conducted following Chilean testing criteria and aimed to confirm if the NMB Splice Sleeve couplers meet Chilean quality requirements while tested using reinforcing bars produced in this country.

2. Laboratory information

SPLICE SLEEVE JAPAN, LTD.
NISSCO TECHNICAL CENTER
313-5 Mukai-machi, Minami-hanashima, Matsudo-shi,
Prefecture of Chiba, Japan

3. Test's details

3.1 Materials that were used in the evaluation tests

Material that were used to assemble the Splice Sleeve test specimens are as shown below.

（1）Reinforcing bars

Manufacturer：GERDAU AZA S.A.
Type：Grade A630-420H (nominal yield strength 420N/mm^2,
nominal tensile strength 630N/mm^2)
Nominal size(cross sectional area)：16mm（201mm^2）、18mm（254mm^2）、22mm（380mm^2）、
25mm（491mm^2）、28mm（615mm^2）、32mm（804mm^2）、
36mm（1,017mm^2）

（2）Couplers

Manufacture: Splice Sleeve Japan, Ltd.
Type and size：NMB Splice Sleeve Type UX（5UX, 6UX, 7UX、8UX、9UX、10UX、11UX）
Coupler's material: FCD700-2 (JIS G 5502)

（3）Filling grout

Manufacturer: Splice Sleeve Japan, Ltd.
Product name：SS Mortar (non-shrink and high-strength grout material)

3.2 Splice Sleeves products details

（1）Couplers

Dimensions and figures along with the material quality of the couplers are shown in Table - 1, Table - 2, and Figure - 1.

Table - 1 NMB Splice Sleeve model UX dimensions (mm)

Sleeve size and type	Rebar size	Sleeve length L (mm)	RS location		Diameters (mm)			Sealing part length (mm)	Minimum rebar embedment (mm)	
			Wide end (L2)	Narrow end (L1)	Outer diameter (A)	Inner diameter				
						Wide end (B)	Narrow end (b)		Wide end	Narrow end
5UX	16mm	245	120	115	45	32	22	15	90	105
6UX	18mm	285	140	135	49	36	25	15	110	125
7UX	20mm	325	160	155	53	40	29	15	130	145
8UX	25mm	370	185	175	58	44	31	15	150	165
9UX	28mm	415	205	200	63	48	35	15	175	190
10UX	32mm	455	225	220	66	51	39	15	195	210
11UX	36mm	495	245	240	71	55	44	15	215	230

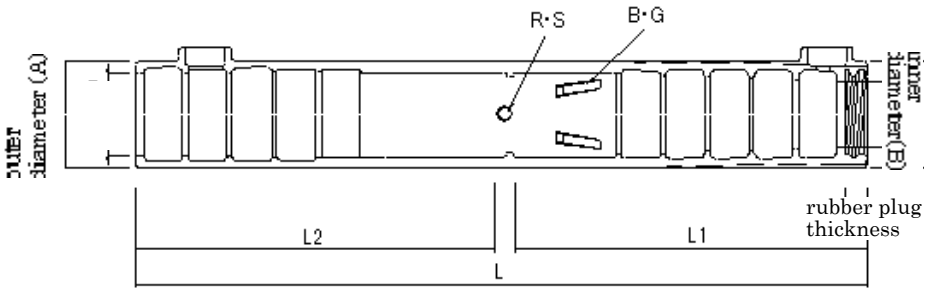


Figure - 1 NMB Splice Sleeve figure UX type

Table - 2 Quality of coupler's material

Coupler's material	Yield point (N/mm^2)	Tensile strength (N/mm^2)	Slip (%)
FCD700-2	420 or higher	700 or higher	2 or higher

3.3 Filling grout
Requirements for SS Mortar are shown below in Table - 3

Table - 3 Filling material quality requirements

Filling grout name	Mixing water (L/bag)	flow test[1] (mm)	Compressive strength (N/mm^2) [2] after curing of 28 days	Expansion[3] (%)
SS Mortar	2.2± 0.1	195± 40	≧80	≧0

Note 1）: Tests were conducted following 「Method of flow test」 as specified in JASS 15 M-103 「Quality requirement of self-leveling materials」. Dimensions of PVC cylinder used in the flow test: inner diameter 50mm, height 100mm.
Note 2）: Test were conducted following ASTM C109 「Standard Test Method for Compressive Strength of Hydraulic Cement Mortars」(using 2-in. or 50mm cubic specimens)
Note 3）: Test conducted following JSCE- F542 「Test method for expansion of mortar」 Specimens were cured during 7 days.

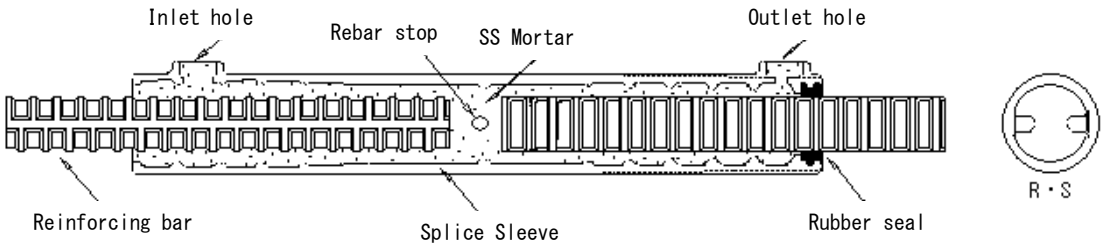
3.4 Coupler's specimens

Table - 4 shows a description of the Splice Sleeve specimens. Figure - 2 is the illustration of the Splice Sleeve specimens.

（1） Tests conducted：Reinforcing bar tensile strength, cyclic tensile tests at elastic and plastic region
（2） Manufacturer：GERDAU AZA S.A.
（3）Reinforcing bar's class/type : NCh204-of2006 「Acero – Barras laminadas en caliente para hormigón armado」 Grade A630-420H (nominal yield strength 420N/mm^2, nominal tensile strength 630N/mm^2)
（4） Nominal diameter：16mm(201mm^2)、18mm(254mm^2)、20mm（380mm^2）、
25mm（491mm^2）、28mm（615mm^2）、32mm（804mm^2）、
36mm（1017mm^2）
（5） Necessary quantity of specimens：5 pieces per size

Table‐4 Specimen's description

Test name	Reinforcing bar manufacturer	Reinforcing bar type/class	Reinforcing bar nominal size	Quantity
Reinforcing bar tensile test	GERDAU AZA S.A.	A630-420H	16mm、18mm、20mm、25mm、28mm、32mm、36mm	5 pieces per size
TYPEⅡ（Cyclic tensile test at elastic and plastic area）	GERDAU AZA S.A.	A630-420H	16mm、18mm、20mm、25mm、28mm、32mm、36mm	5 pieces per size



Figure‐2 Illustration of a NMB Splice Sleeve specimen（example）

3.5 Preparation of Splice Sleeve specimens

Splice Sleeve specimens were prepared following instructions (1) to (7) as shown below. (Marking location and marking length for Splice Sleeve model UX are shown in Table - 5.

（1）Rebar samples were embedded into Splice Sleeve specimens with the minimum embedded length as shown in Table-1.

（2） Splice Sleeve specimens were stood vertically when grouted

（3） Splice Sleeve specimens were grouted using SS Mortar

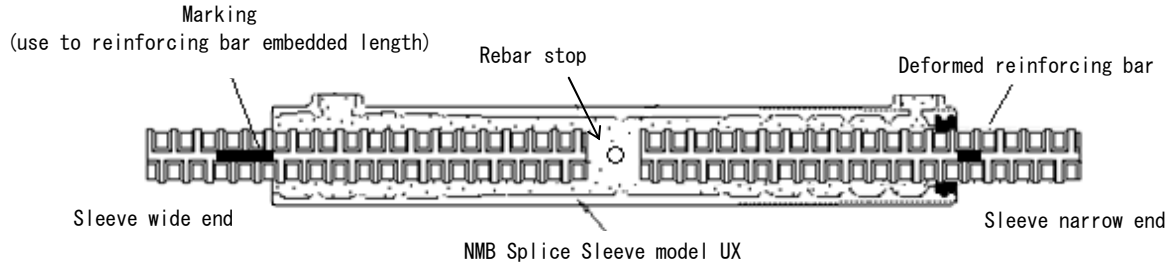
（4）SS Mortar (15kg/bag) was mixed using 2.2 litters ((2.2±0.1 litters) of water per bag. Mixing time was 2 minutes.

（6） Splice Sleeve specimens were cured during one day before removing from the specimen holder.

（7） Specimens of SS Mortar were prepared and tested at the same curing age than the Splice Sleeve specimens.

Table－5 Marking location and length (model UX)

Splice Sleeve size	Bar size	Bar length (mm)	Distance between the bar end and marking (mm)		Length of the marking (mm)	
			Wide end	Narrow end	Wide end	Narrow end
5UX	16mm	600	90	105	30	10
6UX	18mm	600	110	125		
7UX	20mm	650	130	145		
8UX	25mm	650	150	165		
9UX	28mm	700	175	190		
10UX	32mm	700	195	210		
11UX	35mm	700	215	230		



Figure－3 Rebar marking location, Splice Sleeve model UX

3.6 Quality requirement for filling grout material

Flow tests of filling grout were conducted following JASS 15 M-103「Requirements for self-leveling material」(Flow cylinder dimensions : inner diameter 50mm, height 100mm, PVC pipe). Test requirements are shown in Table-6.

Table‐6 Quality requirement of filling grout

Filling grout name	Mixing water (litters/bag)	Temperature of Splice Sleeve (℃)	Water temperature (℃)	SS Mortar temperature (dry) (℃)	Temperature of SS Mortar after mixing (℃)	Flow test (mm)
SS Mortar	2.1～2.3	60 or lower	0～40	－	5～45	195± 40

3.7 Quality requirement for rebar samples

Tests for rebar samples were conducted following NCh 204 「Acero - Barras laminadas en caliente para hormigón」. Requirements are shown in Table-7

Table‐7 Requirement for rebar samples

Rebar class	Yield strength (N/mm^2)	Tensile strength (N/mm^2)	Slip (%)	Tensile strength/Yield strength
A630-420N	430 or higher 580 or lower	630 or higher	8 or higher (700/(tensile strength-K))	1.25 or higher

K is given in the table below. Slip is the distance difference between 2 points which initial distance is L_0=200mm

Reinforcing bar sizes	K
16mm	0
18mm	0
22mm	1.0
25mm	2.0
28mm	3.0
32mm	4.0
36mm	5.0

3.8 Quality requirement for mechanical couplers

The Chilean code NCh430「Hormigón armado - requisites de diseño y cálculos」stablishes that the tests for mechanical couplers shall be conducted following US code AC133「Acceptance criteria for mechanical connector systems for steel reinforcing bars」. The load cycle of cyclic test conducted for mechanical couplers are identical in AC133 (2012) and Japanese codes. Therefore we will adopt Japanese code that requires not only the minimum tensile strength but also certain grade of rigidity, slip resistance and ductility.

NCh430's requirements for mechanical couplers are shown in Table-8. Requirements in Japanese code 「Guideline for reinforcing bar anchors and connectors 2007」 are shown in Table-9, cyclic load of the cyclic test is shown in Figure-4.

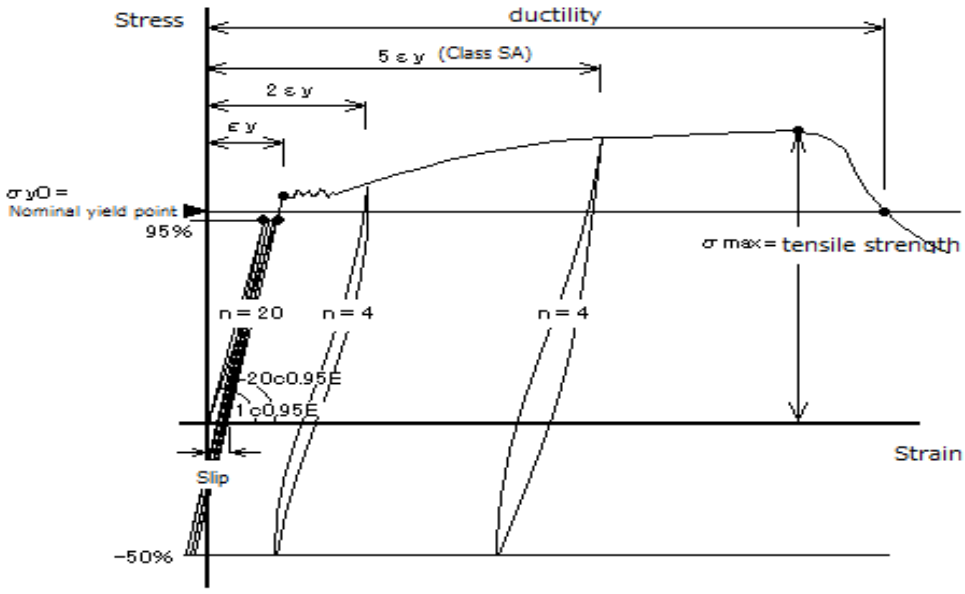


Figure-4 Cyclic load tensile test at elastic and plastic region (same as specified in AC133 (2012) and Japanese codes)

（1）Cyclic tensile test at elastic region

Cyclic load applied repeatedly to a single specimen.

(0.95fyn⇔－0.5fyn） x 20 times

（2）Cyclic tensile test at plastic region

Cyclic load applied repeatedly to a single specimen.

(0.95fyn⇔－0.5fyn） x 20 times

(2ε y⇔－0.5fyn) x 4 times

(5ε y⇔－0.5fyn） x 4 times

Monotonic tension until break or pull out

Splice Sleeve specimen's strain ΔＬ was calculated as follow:

ΔＬ＝ΔＣｍａｘ×（ΔｍＬ÷ΔｃＬ）

Whereas

ΔｍＬ：strain of Splice Sleeve specimen measured by electric strain gage

ΔｃＬ: strain measured from movement of testing machine cylinder

ΔＣｍａｘ: testing machine cylinder deformation measured at rebar failure or measure at fyo after rebar pulling out.

Table‐8 NCh430's Requirements for rebar couplers

Test requirement		Tensile strength (N/mm^2)
Coupler	NCh430	1.40fyo or 1.15fy

fyo: nominal yield strength of reinforcing bar
fy: actual yield strength of reinforcing bar

Table‐9 Requirements for rebar couplers in Japan

		Grade SA	Grade A	Grade B	Grade C
Cyclic load test at elastic area	Strength	fj≧1.35fyn or fun			
	Ductility	εu≧20εy and εu≧0.04	εu≧10εy and εu≧0.02	εu≧5εy and εu≧0.01	-
	Rigidity	E20c≧0.85E1c	E20c≧0.5E1c	E20c≧0.25E1c	-
	Slip	δs(20c)≦0.3mm	δs（20c）≦0.3mm	-	-
Cyclic load test at plastic area	Strength	fj≧1.35fyn or fun		-	-
	Ductility	εu≧20εy and εu≧0.04	εu≧10εy and εu≧0.02	εu≧5εy and εu≧0.01	-
	Slip	εs(4c)≦0.5εy	εs(4c)≦εy	-	-
		δs(4c)≦0.3mm	δs(4c)≦0.6mm		
		εs(8c)≦1.5εy			
		δs(8c)≦0.9mm			

fyn: Reinforcing bar's nominal yield strength
fun: Reinforcing bar's nominal tensile strength
fj: Tensile strength of Sleeve specimen
δs: Slip of Sleeve specimen
Es: Secant stiffness of reinforcing bar at 70% of nominal yield strength
E0.5fyn、E0.7fyn、E0.95fyn:Secant stiffness of Sleeve specimen at 0.5fyn、0.7fyn、0.95fyn
E1c、E20c、E30c: Secant stiffness of Sleeve specimen at 0.95fyn of load cycle No1, No20 and No 30
Es(4c)、εs(8c): Slip strain of Sleeve specimen at load cycle No4 and No8
δ8(4c)、δ8(8c)、δ8(20c)、δ8(30c): Slip of Sleeve specimen at load No4, No8, No20 and No30
εy: Strain at yield point of Sleeve specimen
εu: Strain at ultimate strength point of Sleeve specimen
εs: Slip strain of Sleeve specimen

3.9 Quality requirements for filling grout material

The grout specimens were prepared and tested following ASTM C109「Standard Test Method for Compressive Strength of Hydraulic Cement Mortars」（Using 2-in. or 50-mm cubic specimens）. The specimens were cured submerged into water and at a temperature same as the Splice Sleeve specimens. After curing the specimens were tested at the same age as the Splice Sleeve specimens. Quality requirements for filling grout are shown in Table‐10.

Table‐10 Requirements for filling grout

Grout name	Compressive strength (N/mm^2)
SS Mortar	77 or higher

3.10 Testing devices

Testing machines that were used during the tests are shown in Table-11.

Table‐11 Testing machines details

Device name	Model name	Manufacturer	Material tested	Obtained data
2500kN Fatigue strength testing machine	P10181	Automax	Reinforcing bars and Splice Sleeve specimens	Load and strain
1000kN Compressive strength testing machine	HI－ACTIS	Marui	Filling grout	Load

4. Test results

Performance test results are shown in Table -12, test results of the reiforcing bars are shown in Table - 13, test results of couplers specimens are shown in Tables 14 and 15, test results of filling grout are shown in Table-16.

Table - 12 Test results of SS Mortar

Grout material name	Lot No.	Mixing water Quantity (litters per bag)	Environment temperature (℃)	Water temperature (℃)	Dry grout material temperature (℃)	Grout temperature after mixing (℃)	Flow (mm)
SS Mortar	EL21	2.2	6.0	20.0	13.0	19.0	196×198
Requirement		2.1～2.3	0～60	0～40	—	5～45	195±40

Table - 13 Reinforcing bar samples test results

Reinforcing bar size	No.	Yield Strength (N/mm^2)	Tensile Strength (N/mm^2)	Elongation (%)	Tensile strength/Yield strength
A630-420H 16mm	1	492	679	16.0	1.38
	2	517	702	10.5	1.36
	3	509	699	12.8	1.37
	Average	506	693	13.1	1.37
A630-420H 18mm	1	508	712	13.9	1.4
	2	508	706	13.8	1.39
	3	507	710	15.9	1.4
	Average	508	709	14.5	1.4
A630-420H 22mm	1	449	665	15.8	1.48
	2	451	668	15.8	1.48
	3	454	666	16.6	1.47
	Average	451	666	16.1	1.48
A630-420H 25mm	1	487	696	17.0	1.43
	2	482	691	19.1	1.43
	3	484	694	13.3	1.43
	Average	484	694	16.5	1.43
A630-420H 28mm	1	518	738	18.9	1.42
	2	517	736	17.1	1.42
	3	516	735	18.1	1.42
	Average	517	736	18.0	1.42
A630-420H 32mm	1	499	727	14.8	1.46
	2	500	724	16.3	1.45
	3	500	726	15.3	1.45
	Average	500	726	15.4	1.45
A630-420H 36mm	1	481	703	18.2	1.46
	2	483	699	20.3	1.45
	3	487	703	19.1	1.44
	Average	484	702	19.2	1.45
Requirement		≧420 ≦580	≧630	≧8 and ≧(7,000−tensile strength −K)	≧1.25

Table - 14　Coupler's test results（5UX、6UX、7UX、8UX）

Reinforcing bar	Coupler	No.	Yield Strength (N/mm^2)	Tensile Strength (N/mm^2)	σmax / fy0	σmax / fy	Rigidity($\times 10^5$ N/mm^2)		Rigidity Loss Rate (20C/1C)	Slip (mm)				Ductility(%)	
							0.95σy0E 1c	0.95σy0E 20c		1cδs	20cδs	4cδs (2εy)	8cδs (5εy)	%	εu／εy
A630-420H 16mm	5UX	1	488	669	1.59	1.32	2.63	2.31	0.88	0.06	0.08	0.09	0.19	4.3	24.1
		2	513	696	1.66	1.38	2.51	2.12	0.84	0.06	0.10	0.16	0.31	4.7	24.1
		3	510	690	1.64	1.36	2.65	2.34	0.88	0.05	0.06	0.11	0.22	4.6	24.9
		4	511	695	1.66	1.37	2.88	2.62	0.91	0.01	0.01	0.12	0.20	4.5	26.6
		5	515	700	1.67	1.38	2.78	2.57	0.93	0.03	0.03	0.09	0.19	4.6	25.7
		Average	507	690	1.64	1.36	2.69	2.39	0.89	0.04	0.06	0.11	0.22	4.5	25.1
A630-420H 18mm	6UX	1	515	717	1.71	1.41	2.72	2.47	0.91	0.05	0.05	0.08	0.23	5.4	29.0
		2	511	711	1.69	1.40	2.66	2.45	0.92	0.04	0.04	0.09	0.24	5.3	28.1
		3	476	697	1.66	1.37	2.53	2.16	0.85	0.08	0.11	0.09	0.24	5.1	28.4
		4	503	707	1.68	1.39	2.70	2.49	0.92	0.05	0.05	0.09	0.24	5.6	30.7
		5	507	712	1.70	1.40	2.57	2.32	0.90	0.06	0.06	0.13	0.29	5.5	29.0
		Average	502	709	1.69	1.40	2.64	2.38	0.90	0.06	0.06	0.10	0.25	5.4	29.0
A630-420H 22mm	7UX	1	431	647	1.54	1.43	2.50	2.10	0.84	0.08	0.13	0.08	0.23	4.1	24.5
		2	451	673	1.60	1.49	2.43	1.90	0.78	0.15	0.23	0.08	0.18	5.2	31.8
		3	455	660	1.57	1.46	2.74	2.51	0.92	0.05	0.05	0.09	0.23	3.8	23.4
		4	447	666	1.59	1.48	2.49	2.04	0.82	0.11	0.16	0.11	0.24	5.6	35.1
		5	453	658	1.57	1.46	2.68	2.45	0.92	0.05	0.06	0.08	0.23	3.9	23.6
		Average	447	661	1.57	1.47	2.57	2.20	0.86	0.09	0.13	0.09	0.22	4.5	27.7
A630-420H 25mm	8UX	1	465	668	1.59	1.38	2.48	2.23	0.90	0.07	0.09	0.13	0.33	3.5	18.9
		2	466	666	1.59	1.38	2.49	2.28	0.91	0.06	0.07	0.12	0.33	3.5	19.2
		3	464	672	1.60	1.39	2.49	2.27	0.91	0.07	0.07	0.13	0.34	4.1	22.5
		4	467	658	1.57	1.36	2.52	2.28	0.91	0.07	0.07	0.13	0.33	3.1	16.9
		5	475	658	1.57	1.36	2.57	2.36	0.92	0.05	0.06	0.12	0.30	3.1	17.5
		Average	467	664	1.58	1.37	2.51	2.28	0.91	0.06	0.07	0.13	0.33	3.5	19.0
Required Performance		NCh430			≧1.40	≧1.15	—	—	—	—	—	—	—	—	—
		Class SA			≧1.35	—	≧$0.9E_0$	—	≧0.85	≦0.3	≦0.3	≦0.3	≦0.9	≧4.0	≧20
		Class A			≧1.35	—	≧$0.7E_0$	—	≧0.5	≦0.3	≦0.3	≦0.6	—	≧2.0	≧10

Table - 15　Coupler's test results（9UX、10UX、11UX）

Reinforcing bar	Coupler	No.	Yield Strength (N/mm^2)	Tensile Strength (N/mm^2)	σmax / fy0	σmax / fy	Rigidity（$\times 10^5$ N/mm^2）		Rigidity Loss Rate (20C/1C)	Slip（mm）				Ductility（%）	
							0.95σy0E 1c	0.95σy0E 20c		1cδs	20cδs	4cδs (2εy)	8cδs (5εy)	%	εu／εy
A630-420H 28mm	9UX	1	514	729	1.74	1.41	2.73	2.53	0.93	0.07	0.07	0.11	0.27	4.9	26.6
		2	525	743	1.77	1.44	2.75	2.58	0.94	0.05	0.05	0.11	0.27	5.1	27.4
		3	526	740	1.76	1.43	2.82	2.63	0.93	0.05	0.05	0.11	0.26	4.7	26.0
		4	517	738	1.76	1.43	2.71	2.52	0.93	0.06	0.07	0.10	0.26	4.6	24.7
		5	524	744	1.77	1.44	2.76	2.56	0.93	0.06	0.05	0.12	0.28	5.2	28.3
		Average	521	739	1.76	1.43	2.75	2.56	0.93	0.06	0.06	0.11	0.27	4.9	26.6
A630-420H 32mm	10UX	1	500	704	1.68	1.41	2.57	2.36	0.92	0.07	0.07	0.17	0.44	3.3	17.4
		2	497	701	1.67	1.40	2.58	2.36	0.91	0.06	0.07	0.15	0.42	3.3	17.6
		3	492	708	1.69	1.42	2.55	2.40	0.94	0.06	0.07	0.12	0.29	3.2	16.8
		4	491	702	1.67	1.40	2.54	2.32	0.91	0.06	0.08	0.14	0.38	3.1	16.7
		5	498	695	1.65	1.39	2.57	2.30	0.89	0.06	0.08	0.17	0.43	2.9	15.3
		Average	496	702	1.67	1.40	2.56	2.35	0.91	0.06	0.07	0.15	0.39	3.2	16.8
A630-420H 36mm	11UX	1	476	697	1.66	1.44	2.48	2.31	0.93	0.08	0.08	0.12	0.32	5.1	27.6
		2	480	686	1.63	1.42	2.23	1.90	0.85	0.13	0.16	0.31	0.63	4.0	19.7
		3	476	691	1.65	1.43	2.34	2.02	0.86	0.10	0.14	0.25	0.57	4.5	23.1
		4	483	696	1.66	1.44	2.42	2.22	0.92	0.08	0.09	0.15	0.42	5.2	26.8
		5	478	695	1.65	1.44	2.41	2.13	0.89	0.10	0.11	0.23	0.51	5.0	26.3
		Average	479	693	1.65	1.43	2.38	2.12	0.89	0.10	0.12	0.21	0.49	4.8	24.7
Required Performance		NCh430			≧1.40	≧1.15	—	—	—	—	—	—	—	—	—
		Class SA			≧1.35	—	≧$0.9E_0$	—	≧0.85	≦0.3	≦0.3	≦0.3	≦0.9	≧4.0	≧20
		Class A			≧1.35	—	≧$0.7E_0$	—	≧0.5	≦0.3	≦0.3	≦0.6	—	≧2.0	≧10

Table - 16 Filing grout tensile test result

Curing period	No.	Load (kN)	Compressive Strength (N/mm^2)
28 days	1	299	120
	2	301	120
	3	292	117
	Average	—	119
	Requirement		≧77
*Speciment dimensions 5cm x 5cm x 5cm			

5. Observations

5.1 Reinforcing bars' test results

The pitch of the ribs on the Chilean reinforcing bar samples meet Chilean reinforcing bar requirement.

5.2 Conformity with Chilean coupler requirement (NCh430:2007)

The tensile strength of the grouted couplers NMB Splice Sleeve specimens assembled using Chilean bars was higher than 1.40fy0 and 1.15 fy as required in Chilean Standard NCh430-2007.

5.3 Conformity with Japanese coupler requirement (Guideline for anchors and couplers 2007, Concrete Library No128)

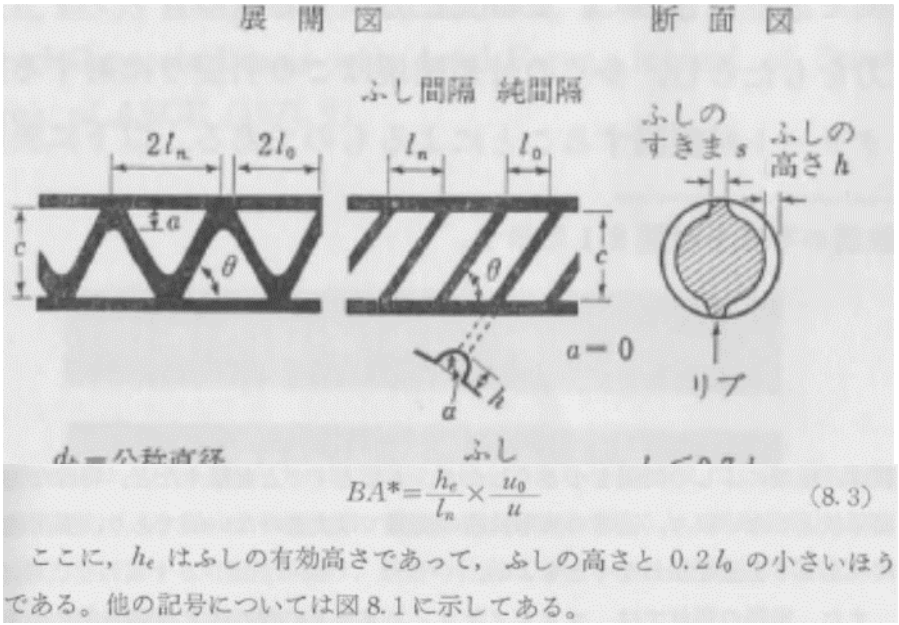
The tensile strength, rigidity loss rate, slip and ductility of coupler's assemblies of 5UX, 6UX, 7UX, 9UX and 11UX meet requirements for Class SA couplers as specified in the Japanese criteria.

The tensile strength, rigidity loss rate, slip of coupler's assemblies of 8UX and 10UX meet requirements specified for Class SA couplers and the ductility meet Class A couplers requirements as specified in the Japanese criteria. As a result, these two sizes are classified as Class A.

The main reason why the ductility of the couplers assemblies was lower than when used to connect commonly used Japanese reinforcing bars, was the reinforcing bar's slip during the test and the pull out. Reinforcing bar pulling out occurred because the pitch distance of the ribs on the Chilean reinforcing bars was smaller than the Japanese bars. Bearing area coefficient of the Chilean bars samples is 0.08 to 0.09 while for the Japanese bars is usually around 0.06. This means that the Chilean bars' bond strength is higher than the Japanese bars. But the distance between ribs in the Chilean reinforcing bars is smaller than the Japanese bars, and this causes a shear slip at the boundary surface of the reinforcing bars. This shear slip constitutes the main cause of the low ductility.

Table - 17 Comparison of the rib distance between Chilean and Japanese deformed bars

Chilean Reinforcing bar (measurment result)	Diameter	16mm	18mm	22mm	25mm	28mm	32mm	36mm
	Distance between 10 ribs (mm)	101.3	102.0	128.8	129.4	166.4	156.5	199.6
Commonly used Japanese Reinforcing bar (measurement resutl)	Diameter	16mm	19mm	22mm	25mm	29mm	32mm	35mm
	Distance between 10 ribs (mm)	107.7	133.2	153.1	173.4	197.2	216.9	241.5
Diference between Chilean and Japanese rebars		−6.4	−31.2	−24.3	−44.0	−30.8	−60.4	−41.9



$$BA^{*}=\frac{h_e}{l_n}\times\frac{u_0}{u} \qquad (8.3)$$

ここに，h_e はふしの有効高さであって，ふしの高さと $0.2l_0$ の小さいほうである。他の記号については図 8.1 に示してある。

Figure - 5 Formula of the bearing area coefficient BA and the components of the deformed reinforcing bar surface.

5.4 Filling grout material

The average of the compressive strength of the filling grout is 119 N/mm^2 and exceeded minimum requirement or 77 N/mm^2.

6. Conclusions

6.1 Conformity with Chilean Standard

The tensile strength of the grouted couplers NMB Splice Sleeve 5UX to 11UX specimens assembled using Chilean bars was higher than 1.40fy0 and 1.15 fy as required in Chilean Standard NCh430-2007.

6.2 Conformity with Japanese Standard

The NMB Splice Sleeve couplers 5UX, 6UX, 7UX, 9UX and 11UX connecting Chilean bars meet requirements for Class SA as specified in the Japanese criteria 「Guideline for reinforcing bar anchors and connectors 2007 」(Japan Society of Civil Engineer Concrete Library 128). For NMB Splice Sleeve couplers 8UX and 10UX the performance meet the requirements of couplers Class A.

In attached document No1 there are shown photos of the Splice Sleeve specimens before testing. Attached document No2 includes photos of the specimens during the tests and attached document No3 shows the graphs of the relation between the stress and strain of the cyclic tensile tests.

Attached Document＃1

Coupler specimen assembly

Photos



Coupler, rebar, grout samples



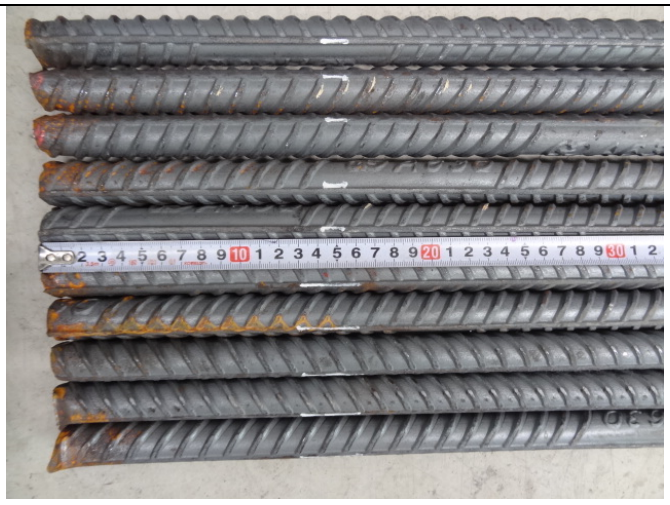
Coupler, rebar, grout samples



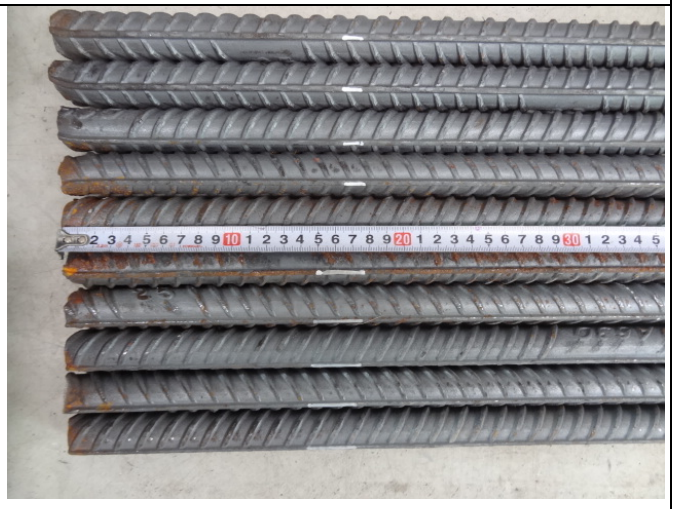
Embedded length control – rebar marking (5UX)



Embedded length control – rebar marking 6UX)



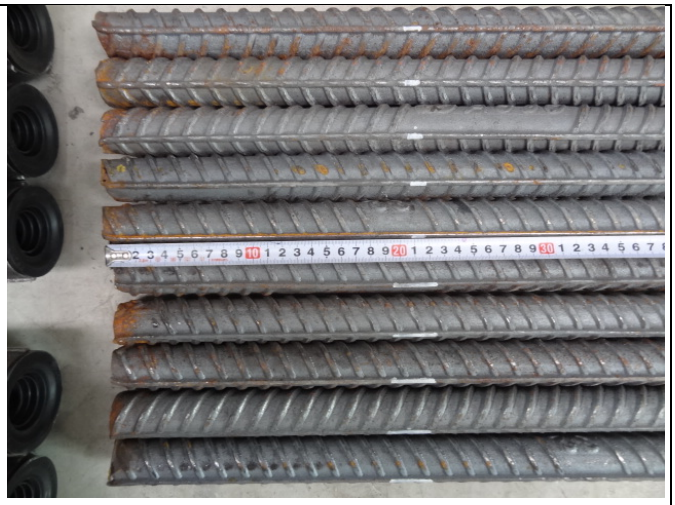
Embedded length control – rebar marking (7UX)



Embedded length control – rebar marking (8UX)



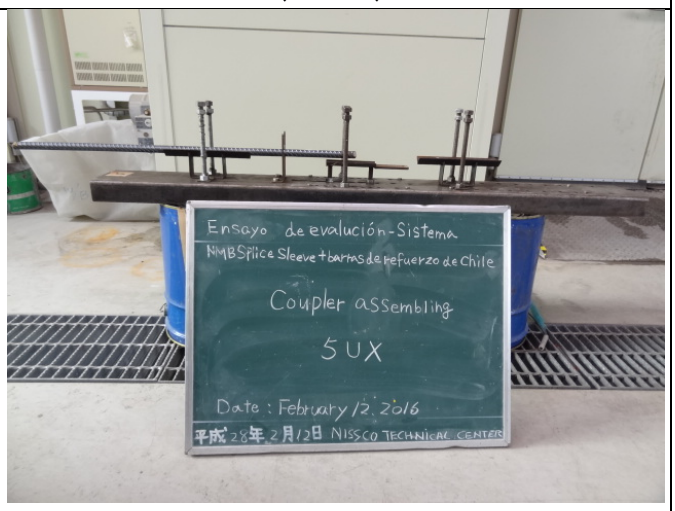
Embedded length control – rebar marking (9UX)



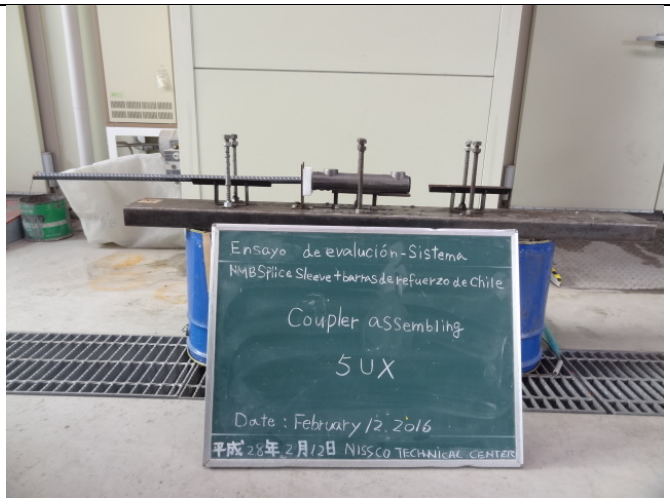
Embedded length control – rebar marking (10UX)



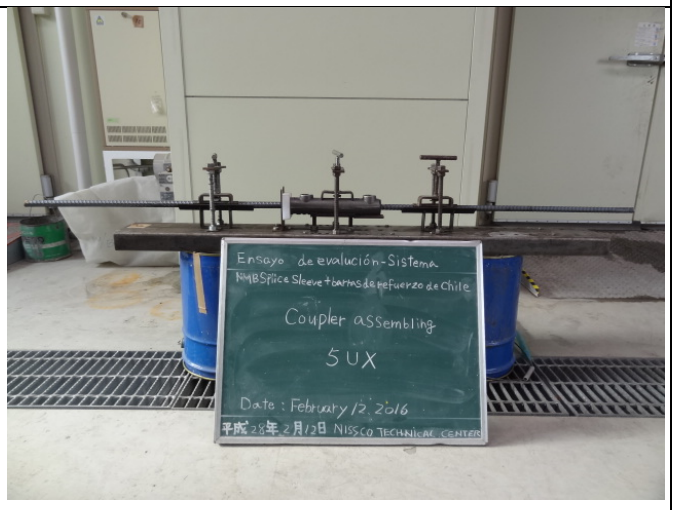
Embedded length control – rebar marking (11UX)



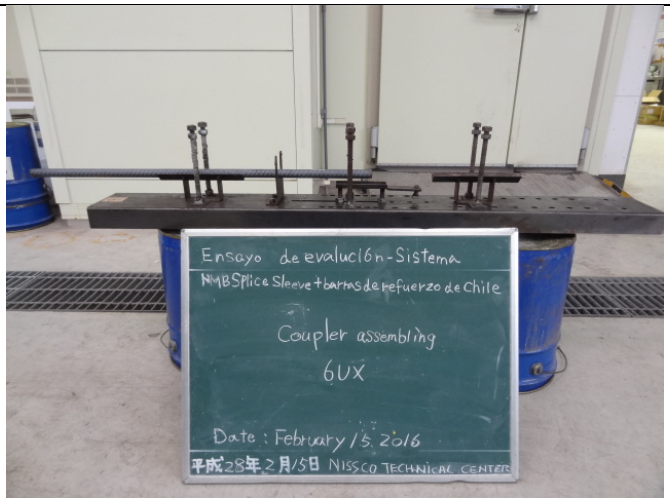
Rebar setting (5UX)



Coupler and rebar setting (5UX)



Coupler and rebar setting (5UX)



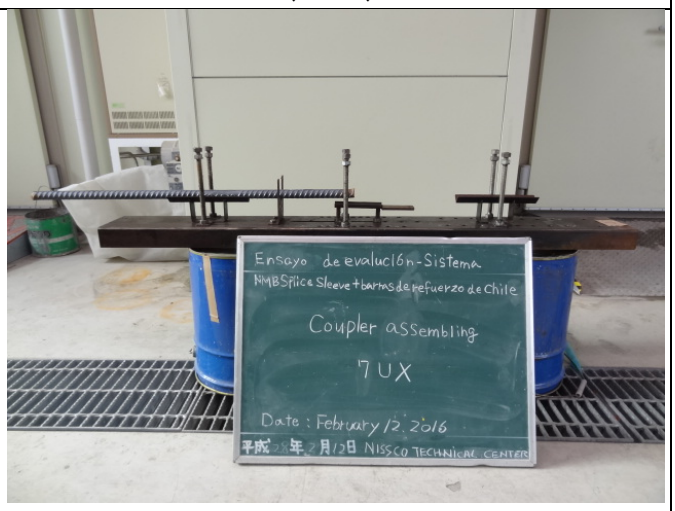
Rebar setting
(6UX)



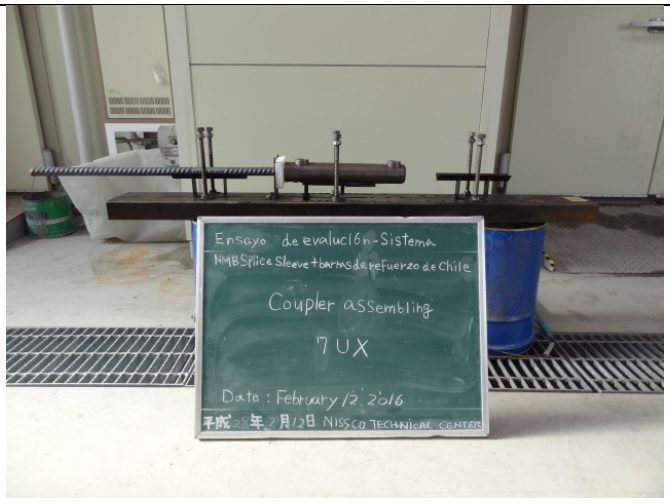
Coupler and rebar setting
(6UX)



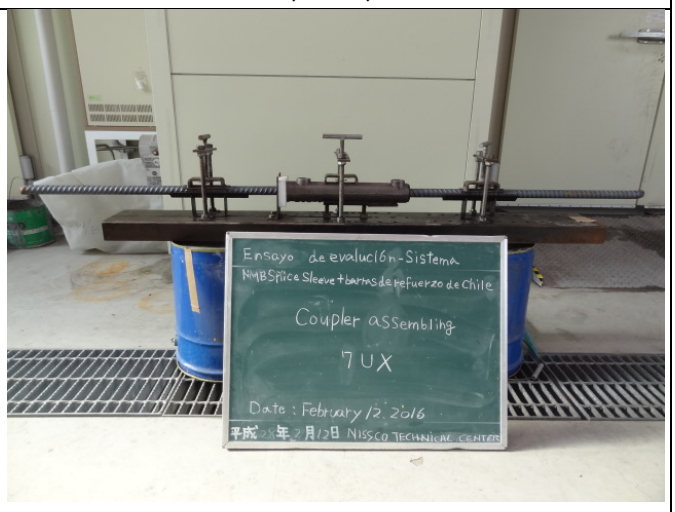
Coupler and rebar setting (6UX)



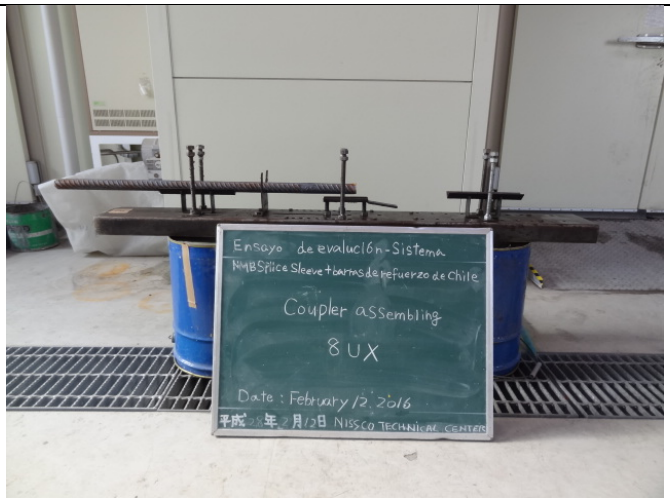
Rebar setting
(7UX)



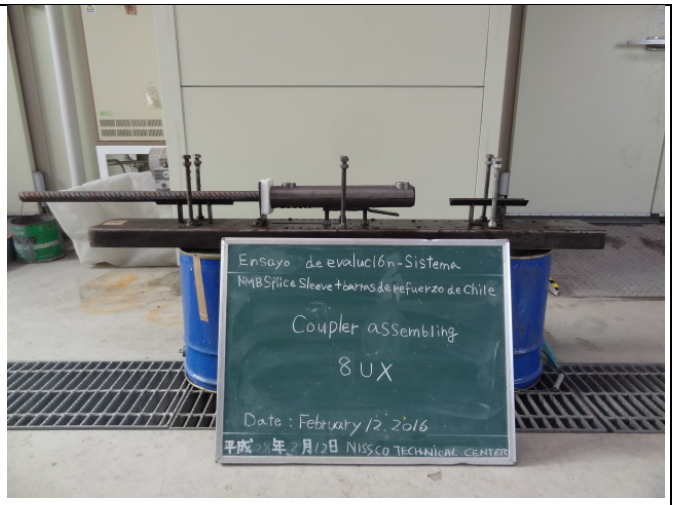
Coupler and rebar setting
(7UX)



Coupler and rebar setting
(7UX)



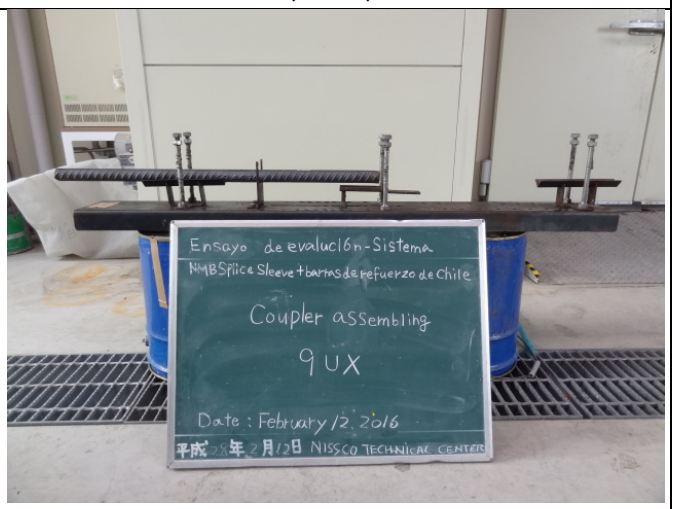
Rebar setting
(8UX)



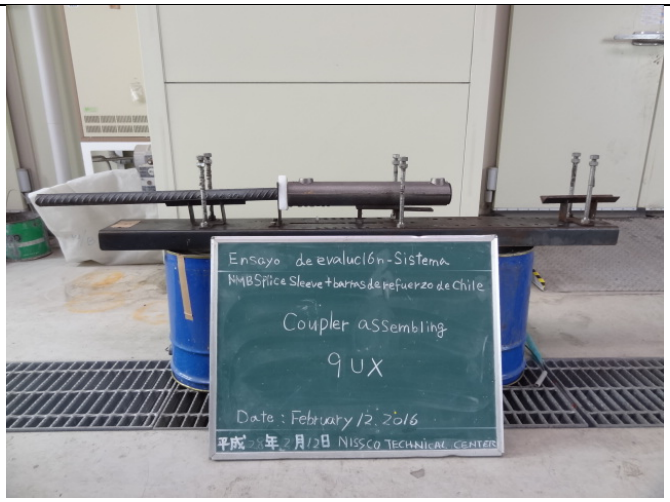
Coupler and rebar setting
(8UX)



Coupler and rebar setting
(8UX)



Rebar setting
(9UX)



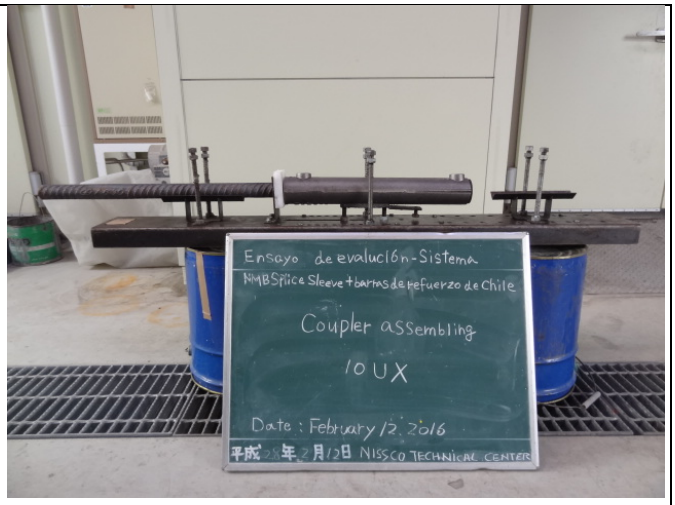
Coupler and rebar setting
(9UX)



Coupler and rebar setting
(9UX)



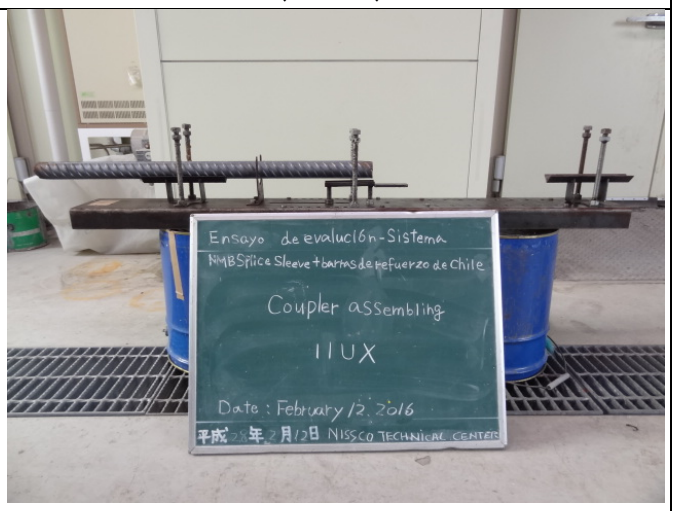
Rebar setting
(10UX)



Coupler and rebar setting
(10UX)



Coupler and rebar setting
(10UX)



Rebar setting
(11UX)



Coupler and rebar setting
(11UX)



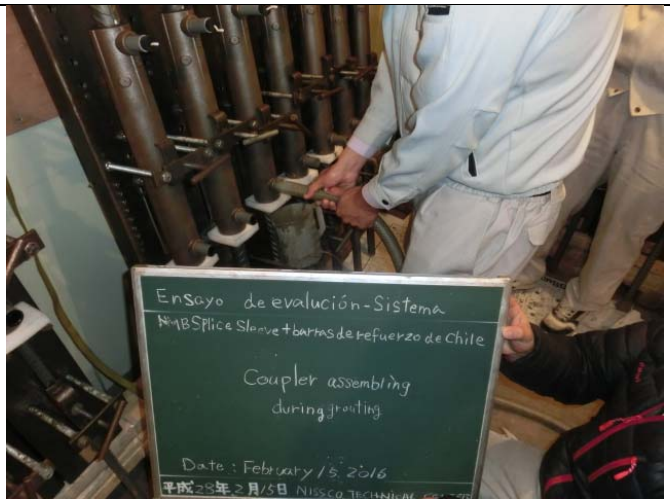
Coupler and rebar setting
(11UX)



All coupler assemblies before grouting



Grout quality control test
(SS Mortar)



Coupler assemblies' grouting



All coupler assemblies after grouting

Attached Document＃2

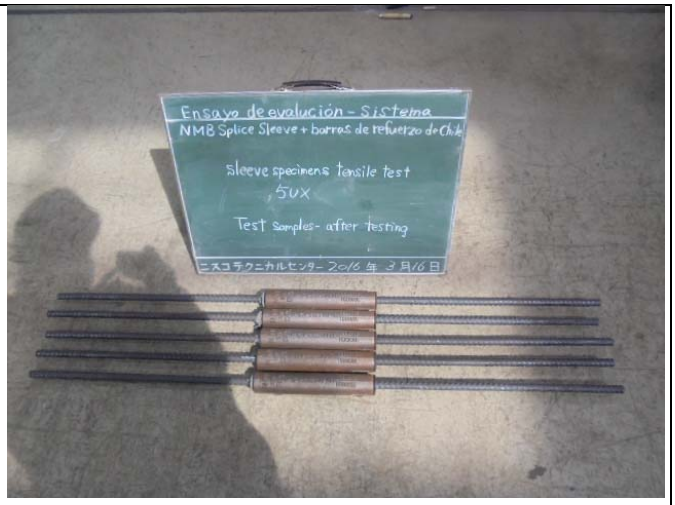
Coupler assemblies test

Photos



Coupler assemblies before testing (No.1～5)

Material used : 16mm 5UX SS Mortar



Coupler assemblies after testing (No.1～5)

Material used : 16mm 5UX SS Mortar



Coupler assembly test (No.1)

Material used : 16mm 5UX SS Mortar



Coupler assembly after testing (No.1)

Material used : 16mm 5UX SS Mortar



Coupler assembly test (No.2)

Material used : 16mm 5UX SS Mortar

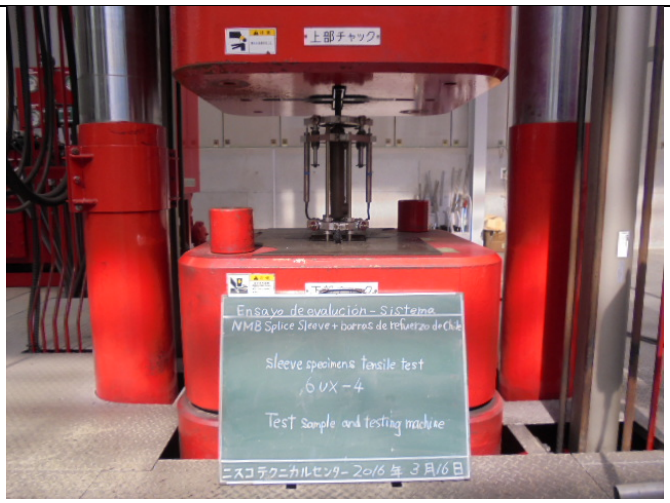


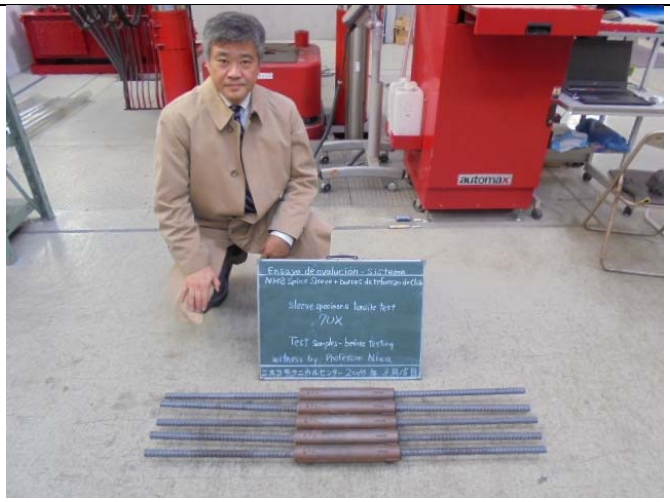
Coupler after testing (No.2)

Material used : 16mm 5UX SS Mortar

	
Coupler assembly test (No.3) Material used : 16mm 5UX SS Mortar	Coupler after testing (No.3) Material used : 16mm 5UX SS Mortar
	
Coupler assembly test (No.4) Material used : 16mm 5UX SS Mortar	Coupler after testing (No.4) Material used : 16mm 5UX SS Mortar
	
Coupler assembly test (No.5) Material used : 16mm 5UX SS Mortar	Coupler after testing (No.5) Material used : 16mm 5UX SS Mortar

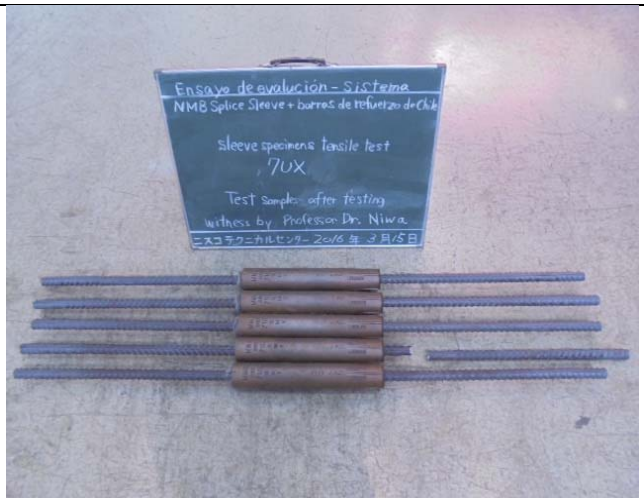
	
Coupler assemblies before testing (No.1～5) Material used : 18mm 6UX SS Mortar	Coupler assemblies after testing (No.1～5) Material used : 18mm 6UX SS Mortar
	
Coupler assembly test (No.1) Material used : 18mm 6UX SS Mortar	Coupler after testing (No.1) Material used : 18mm 6UX SS Mortar
	
Coupler assembly test (No.2) Material used : 18mm 6UX SS Mortar	Coupler after testing (No.2) Material used : 18mm 6UX SS Mortar

	
Coupler assembly test (No.3) Material used : 18mm 6UX SS Mortar	Coupler after testing (No.3) Material used : 18mm 6UX SS Mortar
	
Coupler assembly test (No.4) Material used : 18mm 6UX SS Mortar	Coupler after testing (No.4) Material used : 18mm 6UX SS Mortar
	
Coupler assembly test (No.5) Material used : 18mm 6UX SS Mortar	Coupler after testing (No.5) Material used : 18mm 6UX SS Mortar



Coupler assemblies before testing (No.1～5)

Material used : 22mm 7UX SS Mortar



Coupler assemblies after testing (No.1～5)

Material used : 22mm 7UX SS Mortar



Coupler assembly test (No.1)

Material used : 22mm 7UX SS Mortar



Coupler after testing (No.1)

Material used : 22mm 7UX SS Mortar



Coupler assembly test (No.2)

Material used : 22mm 7UX SS Mortar



Coupler after testing (No.2)

Material used : 22mm 7UX SS Mortar



Coupler assembly test (No.3)

Material used : 22mm 7UX SS Mortar



Coupler after testing (No.3)

Material used : 22mm 7UX SS Mortar



Coupler assembly test (No.4)

Material used : 22mm 7UX SS Mortar



Coupler after testing (No.4)

Material used : 22mm 7UX SS Mortar



Coupler assembly test (No.5)

Material used : 22mm 7UX SS Mortar



Coupler after testing (No.5)

Material used : 22mm 7UX SS Mortar



Coupler assemblies before testing (No.1～5)

Material used : 25mm 8UX SS Mortar



Coupler assemblies after testing (No.1～5)

Material used : 25mm 8UX SS Mortar



Coupler assembly test (No.1)

Material used : 25mm 8UX SS Mortar



Coupler after testing (No.1)

Material used : 25mm 8UX SS Mortar



Coupler assembly test (No.2)

Material used : 25mm 8UX SS Mortar



Coupler after testing (No.2)

Material used : 25mm 8UX SS Mortar

	
Coupler assembly test (No.3) Material used : 25mm 8UX SS Mortar	Coupler after testing (No.3) Material used : 25mm 8UX SS Mortar
	
Coupler assembly test (No.4) Material used : 25mm 8UX SS Mortar	Coupler after testing (No.4) Material used : 25mm 8UX SS Mortar
	
Coupler assembly test (No.5) Material used : 25mm 8UX SS Mortar	Coupler after testing (No.5) Material used : 25mm 8UX SS Mortar



Coupler assemblies before testing (No.1～5)

Material used : 28mm 9UX SS Mortar



Coupler assemblies after testing (No.1～5)

Material used : 28mm 9UX SS Mortar



Coupler assembly test (No.1)

Material used : 28mm 9UX SS Mortar



Coupler after testing (No.1)

Material used : 28mm 9UX SS Mortar



Coupler assembly test (No.2)

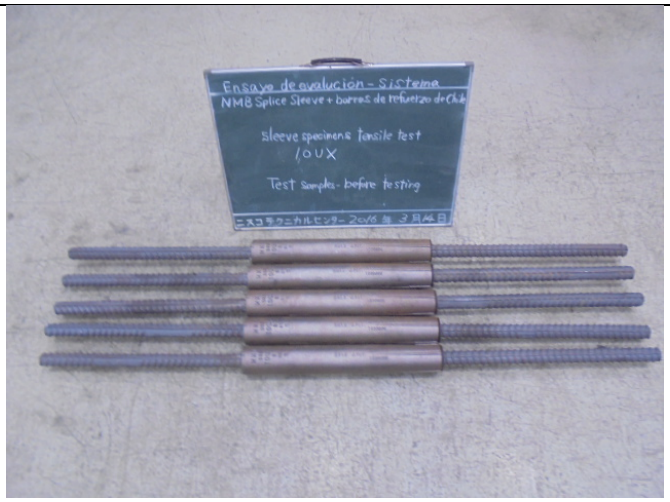
Material used : 28mm 9UX SS Mortar



Coupler after testing (No.2)

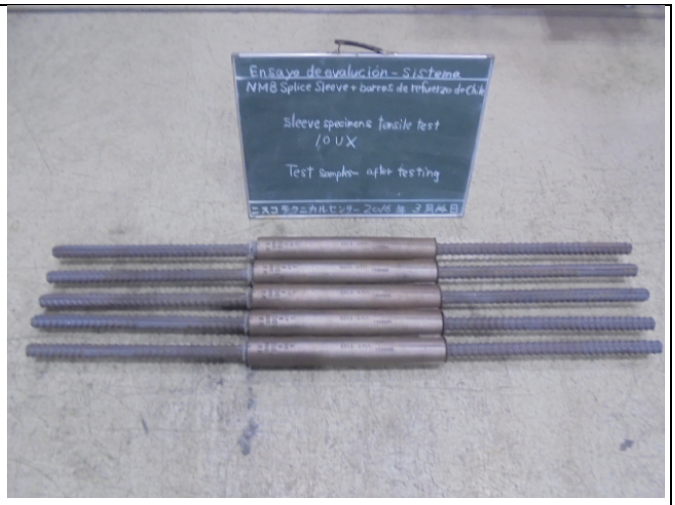
Material used : 28mm 9UX SS Mortar

	
Coupler assembly test (No.3) Material used : 28mm 9UX SS Mortar	Coupler after testing (No.3) Material used : 28mm 9UX SS Mortar
	
Coupler assembly test (No.4) Material used : 28mm 9UX SS Mortar	Coupler after testing (No.4) Material used : 28mm 9UX SS Mortar
	
Coupler assembly test (No.5) Material used : 28mm 9UX SS Mortar	Coupler after testing (No.5) Material used : 28mm 9UX SS Mortar



Coupler assemblies before testing (No.1～5)

Material used : 32mm 10UX SS Mortar



Coupler assemblies after testing (No.1～5)

Material used : 32mm 10UX SS Mortar



Coupler assembly test (No.1)

Material used : 32mm 10UX SS Mortar



Coupler after testing (No.1)

Material used : 32mm 10UX SS Mortar



Coupler assembly test (No.2)

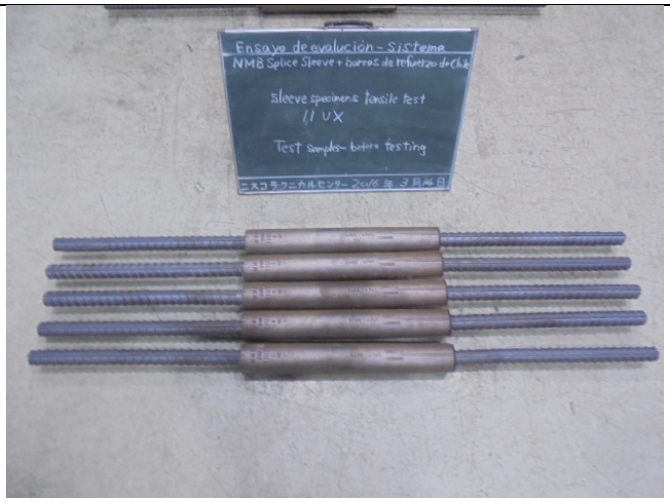
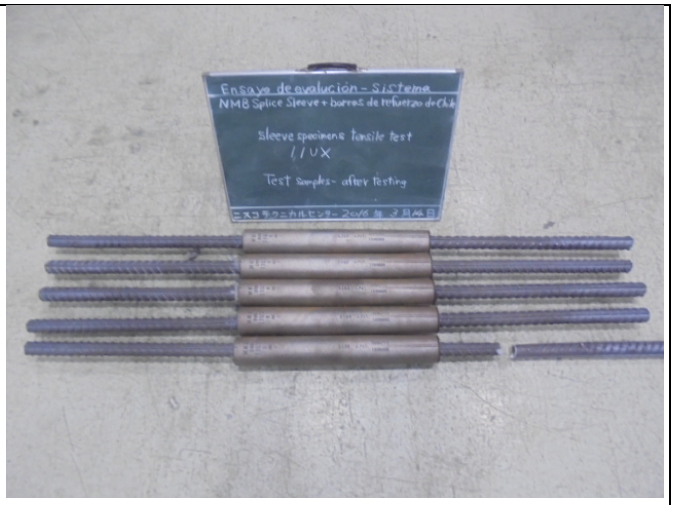
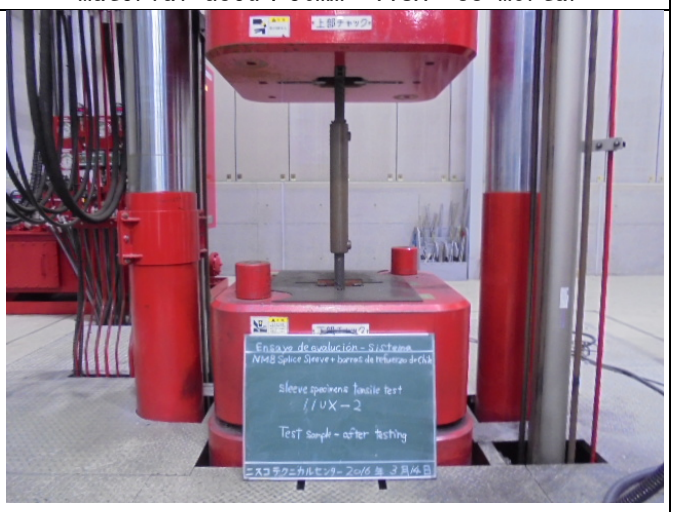
Material used : 32mm 10UX SS Mortar

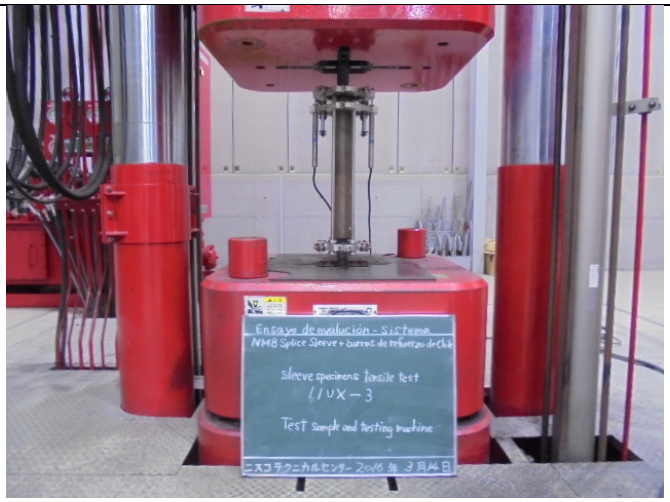


Coupler after testing (No.2)

Material used : 32mm 10UX SS Mortar

	
Coupler assembly test (No.3) Material used : 32mm 10UX SS Mortar	Coupler after testing (No.3) Material used : 32mm 10UX SS Mortar
	
Coupler assembly test (No.4) Material used : 32mm 10UX SS Mortar	Coupler after testing (No.4) Material used : 32mm 10UX SS Mortar
	
Coupler assembly test (No.5) Material used : 32mm 10UX SS Mortar	Coupler after testing (No.5) Material used : 32mm 10UX SS Mortar

	
Coupler assemblies before testing (No.1～5) Material used : 36mm 11UX SS Mortar	Coupler assemblies after testing (No.1～5) Material used : 36mm 11UX SS Mortar
	
Coupler assembly test (No.1) Material used : 36mm 11UX SS Mortar	Coupler after testing (No.1) Material used : 36mm 11UX SS Mortar
	
Coupler assembly test (No.2) Material used : 36mm 11UX SS Mortar	Coupler after testing (No.2) Material used : 36mm 11UX SS Mortar

	
Coupler assembly test (No.3) Material used : 36mm 11UX SS Mortar	Coupler after testing (No.3) Material used : 36mm 11UX SS Mortar
	
Coupler assembly test (No.4) Material used : 36mm 11UX SS Mortar	Coupler after testing (No.4) Material used : 36mm 11UX SS Mortar
	
Coupler assembly test (No.5) Material used : 36mm 11UX SS Mortar	Coupler after testing (No.5) Material used : 36mm 11UX SS Mortar

Grout compression test（After 29days curing、March 15, 2016） No.15



SS Mortar’ s compressive strength test

工事名	Ensayo de evaluación - sistema NMB Splice Sleeve + barras de refuerzo de Chile			
施工箇所	SS Mortar compression test		材料	SS モルタル
採取日	平成28年 2 月 15 日		材齢	29 日
試験日	3 月 15 日	No.	圧縮荷重 (KN)	圧縮強度 (N/mm²)
供試体	立方体 5×5×5cm	1	299	120
		2	301	120
		3	292	117
試験所	ニスコテクニカルセンター		平均	119

SS Mortar test results